

Technical Interview Handbook

Python

Q: What are decorators? A: Functions modifying behavior. Q: GIL? A: Global Interpreter Lock limits parallel bytecode execution in CPython. Q: list vs tuple? A: mutable vs immutable.

SQL

Q: INNER vs LEFT JOIN? A: matching rows vs all left rows. Q: Window functions? A: analytics over partitions. Q: Indexes? A: speed reads, cost writes.

DSA/LeetCode

Practice: arrays, strings, hash maps, two pointers, sliding window, binary search, trees, graphs, heaps, DP, backtracking, union find, tries. Q: BFS vs DFS? A: level vs depth traversal.

System Design

Q: Load balancer? distributes traffic. Cache? reduce latency. DB sharding? horizontal partitioning. CAP theorem basics.

APIs

REST verbs, idempotency, auth(JWT/OAuth), pagination, rate limiting, versioning.

CI/CD

Pipelines, testing, build artifacts, blue-green/canary deploys, rollback.

Behavioral

STAR method, conflict resolution, ownership, debugging stories.